Thai Quoc Bao

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Education

Sai Gon University (SGU – HCM) Expected June 2027
B.Eng. Software Engineering *GPA: 3.0/4.0*

Relevant Coursework

Courses: Object-Oriented Programming, Data Structures & Algorithms, Algorithm Analysis & Design, Database Systems, Operating Systems, Computer Networks, Software Architecture, Discrete Math, Linear Algebra, Probability & Statistics
Certifications: AWS Cloud Practitioner (In Progress), **IELTS Certification (In Progress, Target: 6.5)**

Skills

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, HTML/CSS, LaTeX
Backend & Frameworks: Spring Boot, Spring Security, Hibernate/JPA, Node.js, Express.js, FastAPI, REST APIs
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Docker, GitHub Actions (CI/CD), Nginx, Linux/Bash
Databases & Tools: Neo4j, MySQL, MongoDB, Redis, Git, Postman, Swagger/OpenAPI, IntelliJ IDEA, VS Code

Projects

InteractHub | *ASP.NET Core 8.0, React, Redis, Neo4j, SignalR* Dec. 2025 – Feb. 2026

- Developed a scalable social media platform featuring real-time notifications and hashtag tracking, utilizing WebSockets to replace inefficient polling-based updates
- Architected a robust event pipeline using ASP.NET Core 8.0 and SignalR, integrating **Redis Streams** as a message backbone to ensure reliable, high-throughput event buffering and notification delivery
- Modeled complex user relationships in **Neo4j** to enable high-performance graph queries for friend recommendations, complementing primary PostgreSQL storage for structured user data
- Built a dynamic React frontend with WebSocket integration, ensuring sub-second latency for live feeds, pushed notifications, and real-time hashtag popularity updates
- Containerized the microservice architecture and managed staging environments to validate real-time system performance and scalability

CoRAG (Chain-of-Retrieval) | *Llama-3-8B, Unsloth, LangChain, Python* Apr. 2026 – May 2026

- Fine-tuned a Llama-3-8B model using Unsloth to build a Chain-of-Retrieval (CoRAG) architecture that tightly couples retrieval and generation for higher factuality
- Prepared and curated domain-specific datasets, implemented efficient retrieval indices, and integrated the fine-tuned model into LangChain pipelines for downstream QA tasks
- Evaluated retrieval precision and end-to-end response quality, iterating on prompt and retrieval strategies to improve factual grounding and response relevance

AI-Powered PDF RAG System | *Python, FastAPI, LangChain, ChromaDB, Docker* Sept. 2025 – Nov. 2025

- Developed an advanced Retrieval-Augmented Generation (RAG) architecture allowing users to upload PDF documents and query their contents via natural language, improving relevance and factual grounding for large document collections
- Engineered a document ingestion pipeline that chunks PDFs using recursive text splitting, generates vector embeddings via sentence-transformers, and stores them in ChromaDB for sub-second similarity retrieval
- Implemented core services in Python and FastAPI and orchestrated LangChain-based retrieval + prompt pipelines to produce context-grounded answers while significantly reducing hallucinations
- Built a FastAPI backend with async request handling and streaming responses, achieving p95 query latency under 2 seconds for documents up to 200 pages

School Bus Monitoring & Tracking System | *JavaScript, Socket.io, Python, OpenCV, MongoDB, Docker, REST API* Jan. 2025

- Architected a full-stack monitoring system to automate student attendance tracking across a fleet of 15+ school buses, replacing a manual roll-call process that was error-prone and time-consuming
- Designed a high-concurrency MongoDB schema with optimized indexing and sharding strategies, reducing average query latency when retrieving real-time attendance records for 500+ enrolled students
- Implemented a multi-threaded facial recognition pipeline using Python, OpenCV, and dlib that processes concurrent camera streams and matches detected faces against enrolled embeddings in O(n log n) time
- Developed a Node.js backend using Socket.io and Express with role-based access controls, serving real-time attendance data to a web dashboard and handling 200+ concurrent WebSocket connections
- Containerized the entire microservice stack with Docker Compose, enabling reproducible deployments and reducing environment setup time from hours to under 5 minutes

Automated Cloud Deployment Pipeline | *GitHub Actions, Docker, AWS EC2, AWS ECR, Java, Spring Boot, Bash* June 2025 –

- Designed and implemented an end-to-end CI/CD pipeline using GitHub Actions to automate the build, test, and deployment lifecycle of a Dockerized Spring Boot application to AWS EC2
- Configured a multi-stage Docker build that compiles the Java application with Maven, packages it into a lightweight JRE-based image, and pushes to AWS ECR, reducing final image size by 60% compared to a naive build
- Automated zero-downtime deployments via a custom Bash script that pulls the latest image on EC2, performs health checks, and rolls back to the previous version if the new container fails readiness probes within 30 seconds
- Reduced manual deployment time from 25+ minutes to under 3 minutes per release, enabling the team to ship features with a consistent and repeatable process across staging and production environments

Volunteer & Open-Source Contributions

Open-Source Contributor | *Self-Directed / Volunteer Contributions* Sept. 2024 – Present

Contributing to open-source JavaScript, Python, and tooling projects on GitHub. Focused on writing clean, well-tested backend modules, reviewing pull requests, improving API documentation, and increasing test coverage.

Competitive Programming Club | *Active Member* Sept. 2023 – Present

Solve 150+ algorithmic problems on LeetCode and Codeforces. Regularly practice graph algorithms, dynamic programming, and advanced data structures to sharpen problem-solving speed and Big-O analysis skills.

Peer Tutor – Programming Fundamentals | *Volunteer* 2023 – Present

Mentor 10+ junior students weekly in Java OOP, data structures, and introductory algorithms, improving their assignment pass rates and deepening personal understanding of core CS fundamentals.